

Water Supply — Pressure, Materials and Valves

The supply side is pressurized, so its failures are the opposite of the drain side: not blockage but restriction and corrosion.

The numbers here

Water pressure — normal range	40 to 80 psi
Pressure-reducing valve required above	80 psi
Hot water pressure	12 to 22 psi
Water line at entry to the home — minimum size	3/4 in
Fixture branch line	1/2 in

Three-quarters at the entry so several fixtures can run at once; branches step down to a half inch at the fixture. Above 80 psi the system needs a pressure-reducing valve, because sustained high pressure wears out valves, washers and appliance inlets.

YOU HAVE MISSED THIS

The pressure in a water system will be the greatest when it is:

Located closest to the service entrance to the home.

Pressure drops with every foot of pipe, every fitting and every foot of height climbed. So it is highest where the water comes in and lowest at the furthest, highest fixture — which is where you test if you want to find a problem.

Materials, and why mixing them is a defect

Galvanized steel	Rusts internally, narrowing the diameter. The pipe looks fine from outside while the bore closes up, which is why it is the classic answer to low water pressure in an older house.
Electrolysis	The reaction of copper in direct contact with galvanized piping. Two dissimilar metals in contact with water make a small battery, and one of them corrodes away.
Modern acceptable materials	Copper, PEX and CPVC.
Polybutylene	Grey plastic supply pipe manufactured from 1978 to mid-1995. Its fittings fail, and the manufacturing dates are how you recognize it in a house of that age.
Lead	Lead supply piping or lead solder is a health hazard.

YOU HAVE MISSED THIS 2 TIMES

Low water pressure in the house may be due to water pipes made of:

Galvanized pipe.

YOU HAVE MISSED THIS 2 TIMES

Mixed-use of pipe material in a water distribution piping system is:
Not recommended.

Note the wording of that answer. It is not "a code violation" and not "acceptable" — it is not recommended, because the risk is electrolysis at every junction between the two metals.

The two valves

Gate valve	The main shutoff valve to the property. It is built to sit fully open or fully shut.
Globe valve	A valve that can either adjust or stop the flow of water. It is built to throttle, so it can sit part open.

Say it back

Cover the page. Answer each one out loud.

1. Normal water pressure range, and the threshold for a pressure-reducing valve?
2. Entry line size and fixture branch size?
3. Where in the house is pressure greatest, and where would you test for a problem?
4. Why do old galvanized pipes cause low pressure?
5. When was polybutylene manufactured, and what fails on it?
6. What is electrolysis, and what causes it?
7. Is mixing pipe materials a violation? Give the exact wording.
8. Gate valve or globe valve — which throttles, and which is the main shutoff?